

Group Theory

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Chapter 1

Introduction

1.1 Group Theory Basics

Let p be a prime number. Let $\mathbb{Z}_p = \{[0], [1], \dots, [p-1]\}$ be the set of residue classes modulo p . That is

$$[i] = \{i + kp \mid k \in \mathbb{Z}\},$$

with addition

$$[i] + [j] = [i + j]$$

and multiplication

$$[i] \cdot [j] = [ij]$$

For simplicity, we write $\mathbb{Z}_p = \{0, 1, \dots, p-1\}$. The ring $\mathbb{Z}_p$ is in fact a field. Apart from 0, every element of $\mathbb{Z}_p$ has a multiplicative inverse. Namely, given $a \in \mathbb{Z}_p$ $a \neq 0$, we conclude from the fact that p is prime that $\gcd(a, p) = 1$. This means that there are integers u and v such that $1 = ua + vp$. Read modulo p , this means that u is the multiplicative inverse of a modulo p . We simply write $u = 1/a$. In order to stress the fact that $\mathbb{Z}_p$ has all the properties of a field, the notation $\mathbb{F}_p$ is also used.

The construction of $\mathbb{F}_p$ given just now is not the only way to find a finite field. A second way is to use the polynomial factor ring $\mathbb{F}_p[x]$ modulo $m(x)$ where $m(x)$ is an irreducible polynomial in $\mathbb{F}_p[x]$ of degree $e \geq 2$. The elements of this ring are the residue classes of polynomials

$$[a(x)] = \{a(x) + b(x)m(x) \in \mathbb{F}_p[x] \mid a(x), b(x) \in \mathbb{F}_p[x]\}.$$

Each residue class contains a unique element $c(x)$ satisfying $\deg c(x) < \deg m(x)$. The elements of the field are the residue classes just described. A reduced set of representative of all classes is the set

$$\{[a(x)] \in \mathbb{F}_p[x]/(m(x)), \quad \text{with } \deg a(x) < \deg m(x)\}.$$

The degree condition implies that

$$a(x) = a_{e-1}x^{e-1} + \dots + a_1x + a_0, \quad \text{with } a_i \in \mathbb{F}_p \text{ for all } i = 0, \dots, e-1.$$

Thus the elements are in bijection to the sequences

$$(a_0, a_1, \dots, a_{e-1}) \in \mathbb{F}_p^e$$

and there are p^e such sequences.

Using the letter q to denote a prime p in the first construction or a prime power such as p^e in the second construction, we now have finite fields of order q for any prime power q . In either case, it is common to

denote the field as $\mathbb{F}_q$. If the context is clear, the elements of $\mathbb{F}_q$ are coded as integers. To this end, the mapping

$$\mathbb{F}_p^e \rightarrow \mathbb{Z}_{p^e}, (a_0, a_1, \dots, a_{e-1}) \mapsto \sum_{i=0}^{e-1} a_i p^i$$

is used. If $\mathbb{F}_F$ is any finite field, the prime field is the field that contains 0, 1, $1+1$, $1+1+1$ and so forth. It is true that the prime field of $\mathbb{F}_{p^e}$, p prime, is $\mathbb{F}_p$. The order of the prime field is called the characteristic of $\mathbb{F}$. The prime field is a subfield of $\mathbb{F}$. The set of subfields of $\mathbb{F}_{p^e}$ are exactly the $\mathbb{F}_{p^d}$ for d dividing e . Note that for any subfield $\mathbb{F}_0$ of a field $\mathbb{F}$, the large field is a vector space over the smaller field. In particular, in case that the subfield index is finite, $\mathbb{F}$ has a field basis over the subfield. This is a basis of $\mathbb{F}$ as vector space over $\mathbb{F}_0$. So, for d a divisor of e , the field $\mathbb{F}_{p^e}$ is a vector space of dimension e/d over $\mathbb{F}_p^d$.

Let $V = V(n, q)$ be the vector space of dimension n over the field $\mathbb{F}_q$. Suppose that $q = p^e$ for some prime p . Let ϕ be the Frobenius map from $\mathbb{F}_q$ to $\mathbb{F}_q$ fixing $\mathbb{F}_p$. This means that

$$\phi : x \mapsto x^p.$$

ϕ is an automorphism of $\mathbb{F}_q$ over $\mathbb{F}_p$, which means that

$$(x + y)^\phi = x^\phi + y^\phi, \quad \text{and} \quad (xy)^\phi = x^\phi y^\phi.$$

In fact the automorphism group of $\mathbb{F}_q$ is the cyclic group of order e generated by ϕ , with $\phi^e = \text{id}$. For any divisor d of e , there is a subfield $\mathbb{F}_{p^d}$ of $\mathbb{F}_q$. The subgroup generated by ϕ^d is the automorphism group of $\mathbb{F}_q$ over $\mathbb{F}_{p^d}$. These are the automorphisms of $\mathbb{F}_q$ which fix the elements of $\mathbb{F}_{p^d}$ pointwise.

Let V and W be two vector spaces over a field $\mathbb{F}$. A linear map (or vector space homomorphism) is a function $f : V \rightarrow W$ such that

1. $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in V$, and
2. $f(k\mathbf{x}) = kf(\mathbf{x})$ for all $\mathbf{x} \in V$ and all $k \in \mathbb{F}$.

The kernel (or nullspace) of a linear map is the set

$$N(f) = \{\mathbf{x} \in V \mid f(\mathbf{x}) = \mathbf{0}\}.$$

The dimension of $N(f)$ is called the nullity of f . It is a subspace of V . The image of f (or range of f) is the set

$$R(f) = \{f(\mathbf{x}) \in W \mid \mathbf{x} \in V\}.$$

It is a subspace of W . The dimension of $R(f)$ is called the rank of f . We have that $\dim N(f) + \dim R(f) = \dim V$.

In the case of finite dimensional vector spaces V and W , a linear map can be defined by means of a matrix. Simply pick a basis β for V and γ for W . If, say, $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ then there is a one-to-one correspondence between the linear maps from V to W and the matrices of size $m \times n$ with entries in $\mathbb{F}$. Given a matrix $A = (a_{ij})$, we can define a map that sends $\mathbf{v}_i$ to $\sum_{j=1}^n a_{ij} \mathbf{w}_j$. By linear extension, this defines a map from V to W . If $\mathbf{x}$ is a coordinate vector (assumed to be a row vector) of a vector $\mathbf{v} \in V$, then

$$f(\mathbf{x}) = \mathbf{x}A = \mathbf{y}$$

is the coordinate vector of the image of $\mathbf{x}$ under f in W . This means that

$$\sum_{i=1}^m x_i \mathbf{v}_i \mapsto \sum_{j=1}^n \sum_{i=1}^m x_i a_{ij} \mathbf{w}_j$$

Converseley, given a linear map f , we can associate a matrix by writing the image of a basis β of V in terms of the basis γ of W . Namely, if

$$f(\mathbf{v}_i) = \sum_{j=1}^n \sum_{i=1}^m x_i a_{ij} \mathbf{w}_j,$$

then

$$A = (a_{ij})$$

is a matrix that represents f with respect to the bases β and γ . The assignments $A \mapsto f$ and $f \mapsto A$ are mutually inverse. Thus, in the case of finite dimensional vector spaces, matrices and linear maps are in one-to-one correspondence. If f is an invertible endomorphism, the inverse map is given by the inverse matrix. In the special case that f is the identity endomorphism, the matrix is the identity map.

Let us move on to semi-linear maps. These maps may involve an automorphism of the underlying field. The reason why we consider these maps is that the full automorphism group of the vector space structure leads to these groups.

Let V and W be two vector spaces over a field $\mathbb{F}$. A semi-linear map is a function $f : V \rightarrow W$ such that

1. $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in V$, and
2. $f(k\mathbf{x}) = k^\sigma f(\mathbf{x})$ for some field automorphism $\sigma \in \text{Aut}(\mathbb{F})$ for all $\mathbf{x} \in V$ and all $k \in \mathbb{F}$.

In the second part of the definition, a field automorphism σ arises. This automorphism is the same for all $\mathbf{x} \in V$ and all $k \in \mathbb{F}$. We say that σ is associated with f . Let $\text{SHom}_{\mathbb{F}}(V, W)$ be the set of semilinear $\mathbb{F}$ -homomorphisms from V to W .

Just like with linear maps, we will now define semilinear maps using bases and numerical data. This time, though, we need a matrix $A = (a_{ij})$ and a field automorphism. Since $\text{Aut}(\mathbb{F}_q)$ is a cyclic group, generated by Φ , it suffices to record the integer i in Φ^i to determine the automorphism. We consider i as an element in $\mathbb{Z}_e$. Thus we have a map

$$(\mathbb{F}_q^{n \times m}, \mathbb{Z}_e) \rightarrow \text{SHom}_{\mathbb{F}_q}(V, W), (A = (a_{ij}), h) \mapsto f$$

where

$$f(\mathbf{x}) = (\mathbf{x}A)^{\Phi^h} = \mathbf{y}$$

is the coordinate vector of the image of $\mathbf{x}$ under f in W . This means that

$$\sum_{i=1}^m x_i \mathbf{v}_i \mapsto \sum_{j=1}^n \left(\sum_{i=1}^m x_i a_{ij} \right)^{\Phi^h} \mathbf{w}_j$$

Definition

Let S be a set. A **binary operation** on S is a function $S \times S \rightarrow S$

Examples

The four basic arithmetic operations $(+, -, \cdot, /)$ are operations for many different kinds of sets (integers, rational numbers, real numbers, complex numbers, vectors, matrices – division is not always defined though). The functional composition $\circ$ is a binary operation on the set of functions from a set X to itself. The modulo n arithmetic operations are operations. Matrix multiplication is a binary operation on the set of square matrices of a fixed size. Vector addition and subtraction are binary operations. The cross product of vectors in $\mathbb{R}^3$ is a binary operation.

The following are not binary operations: $\sqrt{x}$, e^x , $\frac{1}{x}$, $A^\top$, $\lim_{x \rightarrow \infty} f(x)$, $\frac{d}{dx}f(x)$, $\int f(x) dx$.

Not all combinations of a binary operation and a set are equally interesting. What makes a binary operation on a set interesting are the group axioms. Depending on the type of operation, these axioms come in different notation. But abstractly, the concept is the same. In order to make clear the different notations, we spell out the definition of a group in the three most common binary operations. The letter G is most commonly used for groups, so we switch from S to G in these definitions. The multiplicative operator $\cdot$ is often omitted. Juxtaposition of symbols implies multiplication. The set S can be a set of functions from a set X to itself. There is a special function called the identity function, and abbreviated as id . It sends every x to itself. So, $\text{id}(x) = x$ for all $x \in X$.

Definition (Group 1)

A binary operation $+$ on a set G is called a group if

1. Associativity: $a + (b + c) = (a + b) + c$ for all a, b, c in G .
2. Existence of Identity: There is an element 0 such that $0 + a = a + 0 = a$ for all $a \in G$.
3. Existence of the inverse element: For each $a \in G$ there is an element $-a \in G$ such that $a + (-a) = (-a) + a = 0$.

Definition (Group 2)

A binary operation $\cdot$ on a set G is called a group if

1. Associativity: $a(bc) = (ab)c$ for all a, b, c in G .
2. Existence of Identity: There is an element 1 such that $1a = a1 = a$ for all $a \in G$.
3. Existence of the inverse element: For each $a \in G$ there is an element $a^{-1} \in G$ such that $a(a^{-1}) = (a^{-1})a = 1$.

Definition (Group 3)

A binary operation $\circ$ on a set of functions G is called a group if

1. Associativity: $a \circ (b \circ c) = (a \circ b) \circ c$ for all a, b, c in G .
2. Existence of Identity: There is an element id such that $\text{id} \circ a = a \circ \text{id} = a$ for all $a \in G$.
3. Existence of the inverse element: For each $a \in G$ there is an element $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = \text{id}$.

Of course, we see that the three seemingly different definitions are all incarnations of the same idea. This is the point of view of abstract algebra. Things that seemingly look different may be observed to have the same structure. This is a very important idea, called isomorphism.

We write $(G, *)$ if we want to emphasize the binary operation $*$ on the set G . Here, $*$ is a placeholder for the actual symbol that may be used. Let $\text{Mat}(m, n, R)$ be the set of $m \times n$ matrices with entries in the ring R . If there is an extra condition $\mathcal{C}$, we write $\text{Mat}(m, n, R; \mathcal{C})$. Let $\mathcal{P}_n(R) = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in R\}$ be the set of polynomials of degree at most n with coefficients in the ring R . For an integer $n \geq 2$, let

$$U_n = \{a \in \mathbb{Z}_n \mid \gcd(a, n) = 1\}$$

be the prime residue classes modulo n .

Examples (Group 1)

Additive groups are:

1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$.
2. $(\text{Mat}(m, n, \mathbb{R}), +)$, $(m, n \in \mathbb{Z}_{\geq 1})$ and $\mathcal{P}_n(\mathbb{R})$ ($n \in \mathbb{Z}_{\geq 0}$),
3. $(\mathbb{Z}_n, +)$ ($n \in \mathbb{Z}_{\geq 2}$),
4. $(\mathbb{R}^n, +)$ ($n \in \mathbb{Z}_{\geq 0}$),

Examples (Group 2)

Multiplicative groups are:

1. $(\mathbb{Q}^+, \cdot)$, $(\mathbb{R}^*, \cdot)$, $(\mathbb{C}^*, \cdot)$,
2. $(\text{Mat}(n, n, \mathbb{R}; \det \neq 0), \cdot)$, $(n \in \mathbb{Z}_{\geq 1})$, $(\text{Mat}(n, n, \mathbb{R}; \det = 1), \cdot)$, $(n \in \mathbb{Z}_{\geq 1})$,
3. $(\mathbb{Z}_p^*, \cdot)$, $(p \in \mathbb{Z} \text{ prime})$,
4. $(U_n, \cdot)$ ($n \in \mathbb{Z}_{\geq 2}$),
5. $(\{e^{\frac{2\pi k}{n}}, k \in \mathbb{Z}\}, \cdot)$, where $n \in \mathbb{Z}_{\geq 1}$.

Examples (Group 3)

Groups of functions w.r.t. composition are:

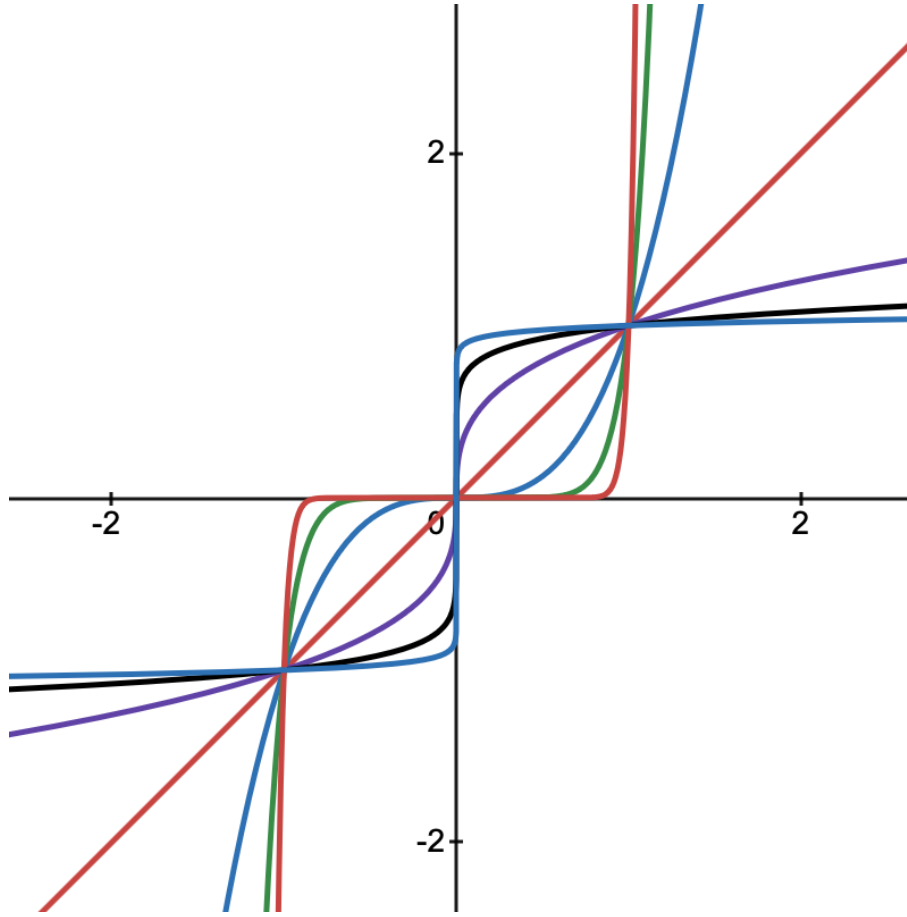
1. $(\text{Sym}(n), \circ)$, $(n \in \mathbb{Z}_{\geq 1})$, $(\text{Alt}(n), \circ)$, $(n \in \mathbb{Z}_{\geq 1})$,
2. $(\{x^{k^n} : \mathbb{R} \rightarrow \mathbb{R} \mid n \in \mathbb{Z}\}, \circ)$, for k an odd integer,
3. $(ax + b : \mathbb{R} \rightarrow \mathbb{R}, a \neq 0, \cdot)$ with $a, b \in \mathbb{R}$, $a \neq 0$.
4. $(\mathbf{x} \mapsto A\mathbf{x}, \circ)$ with $A \in (\text{Mat}(n, n, \mathbb{R}; \det \neq 0))$ and $\mathbf{x} \in \mathbb{R}^n$,
5. $(x^p : \mathbb{F}_q \rightarrow \mathbb{F}_q, \circ)$ where $\mathbb{F}_q$ is the finite field of order q with characteristic p .

Definition

Let G be a group. We say that G is **abelian** if $ab = ba$ for all $a, b \in G$.

Example

Let us take a look at the infinite group $(\{x^{3^n} : \mathbb{R} \rightarrow \mathbb{R} \mid n \in \mathbb{Z}\}, \circ)$. Figure 1.1 shows the graphs of x , x^3 , x^9 , x^{27} , $x^{1/3}$, $x^{1/9}$, $x^{1/27}$. These are just 7 group elements. The identity element is the

Figure 1.1: The group of functions x^{3^n} , $n \in \mathbb{Z}$

function x . We note that $x^3 \circ x^3 = x^9$ for example. The inverse of x^3 (in the sense of functional composition, *not* ordinary multiplication) is $x^{1/3}$, as $x^3 \circ x^{1/3} = x$. The inverse of x^{3^k} is $x^{3^{-k}}$. The set $(\{x^{2^n} : \mathbb{R} \rightarrow \mathbb{R} \mid n \in \mathbb{Z}\}, \circ)$ is not a group as the functions are not one-to-one. This means that the inverse element with respect to functional composition is not defined.

Definition

Let H be a subset of a group G . We say that H is a subgroup of G if H is a group by itself under the restricted binary operation of G . In this case, we write $H \leq G$.

Comments

It is clear that G itself counts as a subgroup of itself. Likewise, the identity element alone is always a subgroup. These two are called the trivial subgroups. The identity element of a subgroup is the same as the identity element of a group. The inverse of an element in the subgroup is the same as the inverse of that element in the group.

Example

Some subgroups of additive groups are:

1. $G = (\mathbb{Z}, +)$, $H = (k\mathbb{Z}, +)$ for $k \in \mathbb{Z}$.
2. $(\text{Mat}(n, n, \mathbb{R}), +)$, $(n \in \mathbb{Z}_{\geq 1})$ and $H = (\{A \in \text{Mat}(n, n, \mathbb{R}) \mid A^\top = A\}, +)$.
3. $(\mathbb{Z}_n, +)$ $(n \in \mathbb{Z}_{\geq 2})$, and $H = (d\mathbb{Z}, +)$ for d a divisor of n .
4. $(\mathbb{R}^n, +)$ $(n \in \mathbb{Z}_{\geq 0})$, $H = (\{\mathbf{x} \in \mathbb{R}^n \mid \sum_{i=1}^n x_i = 0\}, +)$

Example

Some subgroups of multiplicative groups are:

1. $G = (\mathbb{C}^*, \cdot)$, $H = (\{z \in \mathbb{C}^* \mid |z| = 1\}, \cdot)$ (the complex unit circle).
2. $(\text{Mat}(2, 2, \mathbb{R}; \det \neq 0), \cdot)$, $H = \left(\left\{ \begin{bmatrix} \cos(\omega) & -\sin(\omega) \\ \sin(\omega) & \cos(\omega) \end{bmatrix} \mid \omega \in \mathbb{R}, 0 \leq \omega < 2\pi \right\}, \cdot \right)$ (the group of rotations around the origin).

Definition

Let G be a group. Let S be a subset of G . Let $\langle S \rangle$ denote the smallest subgroup of G containing S . This subgroup is called the subgroup generated by S . If S is empty, then $\langle S \rangle$ is the identity subgroup of G . If $S = \{s\}$ consists of one element only, we write $\langle s \rangle$ for $\langle S \rangle$. A generating set for G is a set S such that $\langle S \rangle = G$.

Comments

If $S = \{s\}$ consists of one element only, then $\langle S \rangle$ is called cyclic. In this case, $\langle S \rangle = \{s^k \mid k \in \mathbb{Z}\}$. A group G is cyclic if there is an element $g \in G$ such that $\langle g \rangle = G$.

Example

Let p be a prime number. The group $U_p = \mathbb{Z}_p^*$ (with multiplication modulo p) is cyclic (proof omitted). A generator is called a primitive element. While it is known that a generator exists, no easy formula is known to express one in terms of p . However, there are probabilistic algorithms for finding primitive elements in these groups. We can also determine the number of primitive elements, assuming we can factor $p - 1$.

Example

A vector space is an abelian group. A basis is a minimal generating set. The minimal number of generators of a vector space V is the dimension of V .

Definition

Let G be an additive group. An element of g is said to have order n if $ng = g + g + \cdots + g = 0$ (n terms) and $kg \neq 0$ for all k with $0 < k < n$. For a multiplicative group G , the condition is rephrased as $g^n = 1$ and $g^k \neq 1$ for all k with $0 < k < n$. In both cases, we write $|g| = n$. If no such n can be found, then we say that g has infinite order. We write $|g| < \infty$ to express the fact that g has finite order.

Comments

Let G be an additive group. If $|g| = n$ then $\langle g \rangle = \{kg \mid k = 0, \dots, n-1\}$. Let G be a multiplicative group. If $|g| = n$ then $\langle g \rangle = \{g^k \mid k = 0, \dots, n-1\}$. Thus we can say that for an element g of finite order, the order is the size of the subgroup generated by it. In other words, $|g| = |\langle g \rangle|$ if $|g| < \infty$.

It is possible to create new groups from old. One such construction is the direct product:

Definition

Let $(G, *)$ and $(H, \diamond)$ be groups. The direct product of G and H , denoted $G \times H$ is the cartesian product $\{(g, h) \mid g \in G, h \in H\}$ together with the binary operation

$$(g_1, h_1) \cdot (g_2, h_2) := (g_1 * g_2, h_1 \diamond h_2).$$

It is easy to check that the direct product is again a group. For instance, the identity of $G \times H$ is (e_G, e_H) , with e_G the identity in G and e_H the identity in H . Also, the inverse of $(g, h) \in G \times H$ is (g^{-1}, h^{-1}) , which exists in $G \times H$ because g^{-1} exists in G and h^{-1} exists in H . If $|G| < \infty$ and $|H| < \infty$, the order of $G \times H$ is $|G| \cdot |H|$.

The direct product can arise in surprising ways. For instance, there is a way of viewing $\mathbb{Z}_{21}$ as $\mathbb{Z}_7 \times \mathbb{Z}_3$. In order to make this statement precise, we will come back to the question of when different groups can be considered equal. The first step is to look for structure preserving maps. These kinds of maps are called (group-)homomorphisms:

Definition

Let $(G, *)$ and $(H, \diamond)$ be groups. A homomorphism from G to H is a map $f : G \rightarrow H$ such that

$$f(a * b) = f(a) \diamond f(b), \quad \text{for all } a, b \in G.$$

In this case, we say that H is a homomorphic image of G . We define the kernel of f to be

$$\text{Ker } f = \{g \in G \mid f(g) = e_H\}$$

where e_H is the identity element of H . We define the image of f to be

$$\text{Image } f = \{f(g) \mid g \in G\} = f(G).$$

Example

Fix an integer $n \in \mathbb{Z}_{\geq 1}$. For $a \in \mathbb{Z}$ there exist unique integers q and r such that

$$a = qn + r. \quad (1.1)$$

Define a function $\nu_n : \mathbb{Z} \rightarrow \mathbb{Z}_n$ with $\nu_n(a) = r$ with r as in (1.1). Suppose that $a = q_1n + r_1$ and $b = q_2n + r_2$ with $0 \leq r_i < n$. Then

$$a + b = q_1n + r_1 + q_2n + r_2 = (q_1 + q_2)n + (r_1 + r_2).$$

Define $r_3 = r_1 + r_2$ if $r_1 + r_2 < n$ and $r_3 = r_1 + r_2 - n$ otherwise. Then

$$a + b = (q_1 + q_2)n + r_1 + r_2$$

in the first case and

$$a + b = (q_1 + q_2 + 1)n + (r_1 + r_2 - n) = (q_1 + q_2 + 1)n + r_3.$$

with $0 \leq r_3 < n$ in both cases. Therefore, in both cases

$$\nu_n(a + b) = r_3 \equiv r_1 + r_2 = \nu_n(a) + \nu_n(b) \pmod{n},$$

which means that ν_n is a homomorphism from $G = (\mathbb{Z}, +)$ to $H = (\mathbb{Z}_n, +)$. Note that we use the addition in $\mathbb{Z}_n$ when evaluating $\nu_n(a) + \nu_n(b)$, indicated by the $\equiv$ symbol. This homomorphism is known as the **canonical residue class map modulo n** .

Before we proceed, it is convenient to introduce a notation which extends a binary operation to set. Let S and T be subsets of a group $(G, *)$. Then we put

$$S * T = \{s * t \mid s \in S, t \in T\}.$$

In case that $S = \{s\}$ is singleton, we may write

$$s * T = \{s * t \mid t \in T\}.$$

Likewise, if $T = \{t\}$, we write

$$S * t = \{s * t \mid s \in S\}.$$

In the latter two cases, under the additional condition that S and T are subgroups, we call the resulting set a coset. For instance, $s * T$ is a left coset of T (in G), whereas $S * t$ is a right coset of S (in G).

Theorem

Let $(G, *)$ and $(H, \diamond)$ be groups and let $f : G \rightarrow H$ be a group homomorphism from G to H . Then:

1. $f(e_G) = e_H$ where e_G is the identity element in G and e_H is the identity element in H .
2. $f(g^n) = f(g)^n$ for all $g \in G$ and all $n \in \mathbb{Z}$. Here, $g^n = g * g * \cdots * g$ (n times) and $f(g)^n = f(g) \diamond f(g) \diamond \cdots \diamond f(g)$ (n times).
3. If $|g|$ is finite, then $|f(g)|$ divides $|g|$.
4. $\text{Ker } f$ is a subgroup of G .
5. $\text{Image } f$ is a subgroup of H .
6. $f(a) = f(b)$ if and only if $a * \text{Ker } f = b * \text{Ker } f$.

7. If $f(g) = h$, then $f^{-1}(h) = \{a \in G \mid f(a) = h\} = g * \text{Ker } f$.

Theorem

Let G be a group and let U be a subgroup of G . Then

1. A coset of U is a subgroup if and only if it contains the identity element of G .
2. Two cosets of U are either disjoint or equal.
3. G is partitioned into left cosets of U .
4. G is partitioned into right cosets of U .
5. If $|G| < \infty$ then $|U|$ divides $|G|$. (Theorem of Lagrange)
6. If $|G| < \infty$, the number of cosets of U in G is $|G|/|U|$. This number is called the index of U in G .

Examples

1. Let G be a matrix group over the field $\mathbb{F}$. Consider the determinant map $\det : G \rightarrow \mathbb{F}, A \mapsto \det(A)$. Since $\det(AB) = \det(A)\det(B)$, the map is a homomorphism. It also follows that

$$\det(I) = 1, \quad \text{and} \quad \det(A^{-1}) = (\det(A))^{-1}$$

The kernel is the group of matrices of determinant one. If $G = \text{GL}(n, \mathbb{F})$, this subgroup is called the special linear group, or $\text{PSL}(n, \mathbb{F})$. If $\mathbb{F} = \mathbb{F}_q$, we write $\text{PSL}(n, q)$.

2. Neither the map $A \mapsto A^\top$ nor the map $A \mapsto A^{-1}$ are homomorphisms in general. The problem is that $(AB)^\top = B^\top A^\top$ which is not always equal to $A^\top B^\top$. Likewise, $(AB)^{-1} = B^{-1}A^{-1} \neq A^{-1}B^{-1}$. However, in the event that G is abelian (commutative), then both maps are indeed homomorphisms.

Example

Two important homomorphism arises by means of the conjugation map. The first is the conjugation map itself. It exists for every group element. Let G be a group and let $c \in G$. Then

$$f_c : G \rightarrow G, \quad g \mapsto cgc^{-1}$$

is the conjugation by c . The map is onto: given $h \in G$ we put $g = c^{-1}hc$ and observe that $f_c(g) = h$; one-to-one: Assuming $cgc^{-1} = chc^{-1}$ for $g, h \in G$ we conclude that $g = h$; and a homomorphism:

$$f_c(gh) = c(gh)c^{-1} = cghc^{-1} = cgc^{-1}chc^{-1} = f_c(g)f_c(h).$$

The kernel is trivial:

$$\text{Ker } f = \{g \in G \mid cgc^{-1} = e\} = \{g \in G \mid g = c^{-1}c = e\}.$$

Thus $f_c \in \text{Aut}(G)$, the automorphism group of G . Moreover, the map $\iota : G \rightarrow \text{Aut}(G), c \mapsto f_c$ is yet another homomorphism. This is because

$$f_{cd}(g) = (cd)g(cd)^{-1} = c(dgd^{-1})c^{-1} = (f_c \circ f_d)(g)$$

for all $g \in G$, so

$$\iota(cd) = f_{cd} = f_c \circ f_d = \iota(c) \circ \iota(d).$$

The kernel of ι is the set of $c \in G$ such that $\iota(c) = \text{id}$, or equivalently $f_c(g) = g$ or $cgc^{-1} = g$ or $cg = gc$, which are those $c \in G$ which commute with every element of G . This subgroup is called the center of G . The range of ι is a subgroup of $\text{Aut}(G)$ called the **inner automorphism group** of G . It is denoted as $\text{Inn}(G)$. Every $f_c \in \text{Aut}(G)$ is called an inner automorphism of G . Not all automorphisms are inner. Some groups, for instance $\text{Sym}(6)$, have automorphisms that are not inner.

Definition (Left Action)

Let G be a group. Let X be a set. We say that G acts on X (from the left) if there is a map

$$G \times X \rightarrow X, (g, x) \mapsto g \cdot x$$

such that

1. $e \cdot x = x$ for all $x \in X$ and
2. $fg \cdot x = f \cdot (g \cdot x)$ for all $f, g \in G$ and all $x \in X$.

Definition (Right Action)

Let G be a group. Let X be a set. We say that G acts on X (from the right) if there is a map

$$X \times G \rightarrow X, (x, g) \mapsto x \cdot g$$

such that

1. $x \cdot e = x$ for all $x \in X$ and
2. $x \cdot fg = (x \cdot f) \cdot g$ for all $f, g \in G$ and all $x \in X$.

Comments

Let the group G act on the set X from the left. Let $x \in X$. The orbit of x under G is $G(x) = \{g \cdot x \mid g \in G\}$. Let $x \in X$. The stabilizer of x in G is the set $G_x = \{g \in G \mid g \cdot x = x\}$. Two orbits are either the same or disjoint: $G(x) \cap G(y) \neq \emptyset$ implies that $G(x) = G(y)$. A transversal of the orbits of G on X is a set $\{x_1, x_2, \dots, x_\ell\}$ such that $X = \bigcup_{i=1}^\ell G(x_i)$ as a disjoint union. If $y \in G(x)$ with $g \cdot x = y$ then $G_y = gG_xg^{-1}$.

Definition

Let G be a group. Two elements $g, h \in G$ are said to be conjugate if there is an element $c \in G$ such that $f_c(g) = h$. Conjugacy is an equivalence relation on G . The classes are called conjugacy classes of G . The elements in the center of G lie in classes by themselves. A group is abelian if every element forms a class of its own. The conjugacy classes of G are the orbits of $\text{Inn}(G)$ in the natural action on G .

Definition

Let G act on X . Then G acts on subsets of X by means of

$$(g, S) \mapsto g \cdot S = \{g \cdot s \mid s \in S\}.$$

This is the induced action on subsets. Likewise, there are induced actions on sequences, on pairs, and on set partitions of X . If $X = G$ itself and the action is by conjugation, there is an induced action on subgroups of G . Two subgroups are conjugate if they lie in the same orbit. So, if U and V are subgroups of G , then U and V are conjugate if there is an element $g \in G$ such that $gUg^{-1} = V$. Conjugation is an equivalence relation on the set of subgroups. The orbits of this action are called conjugacy classes of subgroups. A subgroup S is called normal if the conjugacy class of S contains only S . In this case, we write $S \trianglelefteq G$. The stabilizer of a subgroup S in the action of G by conjugation is called the normalizer of S , denoted $N(S)$. The group S is normal in its normalizer. A normal subgroup consists of whole conjugacy classes only. That is, if N is normal in G and $\mathcal{C}$ is a conjugacy class of elements of G with $N \cap \mathcal{C} \neq \emptyset$, then $\mathcal{C} \subseteq N$. The identity subgroup and the group G itself are trivial normal subgroups. If N is normal in G , we write $N \trianglelefteq G$.

Remark

The kernel of a homomorphism $f : (G, *) \rightarrow (H, \diamond)$ is a normal subgroup. Indeed, if $h \in \text{Ker } f$ then

$$f(g^{-1} * h * g) = f(g^{-1}) \diamond f(h) \diamond f(g) = f(g)^{-1} \diamond e_H \diamond f(g) = f(g)^{-1} \diamond f(g) = e_H,$$

which shows that

$$g^{-1} * h * g \in \text{Ker } f$$

also. The last condition means that $\text{Ker } f$ is invariant under the conjugation map conj_g for any $g \in G$. Since this is true for any $h \in \text{Ker } f$ we conclude that $\text{Ker } f$ is a disjoint union of conjugacy classes, and hence $\text{conj}_g(\text{Ker } f) = \text{Ker } f$ for any $g \in G$, which means $\text{Ker } f$ is normal in G , or $\text{Ker } f \trianglelefteq G$. Thus, $\text{Ker } f$ is its own conjugacy class of subgroups.

The general linear group is the most important group that arises in the field of linear algebra. It is the group of endomorphisms of a vector space. That means, the elements of the group are invertible linear maps from the space to itself. The group is finite if and only if the underlying field is finite. In the following, we will look at the finite case of these groups.

Let $\text{GL}(n, q)$ be the set of $n \times n$ matrices with entries drawn from $\mathbb{F}_q$ and with determinant nonzero.

$\text{GL}(n, q)$ is a group.

The order of $\text{GL}(n, q)$ is

$$\prod_{i=0}^{n-1} (q^n - q^i).$$

Proof: Consider the number of ways that rows of a matrix can be chosen so that the resulting matrix is invertible. For the first row ($i = 0$), we can pick all but the zero vector, for a total of $q^n - q_0$ possibilities. The remaining rows must be chosen with respect to the choice of the rows already picked. However, the number of possibilities for a given row is always the same. This means we can multiply the numbers. So, for the second row, we can pick any vector except a scalar multiple of the first row. This leaves $q^n - q^1$ possible choices. In general, after i rows have been selected, we have q^i forbidden vectors (the span of the i vectors), leaving $q^n - q^i$ valid choices for the next row. Multiplying these numbers leads to the stated number.

The group $\text{GL}(2, 3)$ has 48 ($= 8 \cdot 6$) elements:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \right.$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

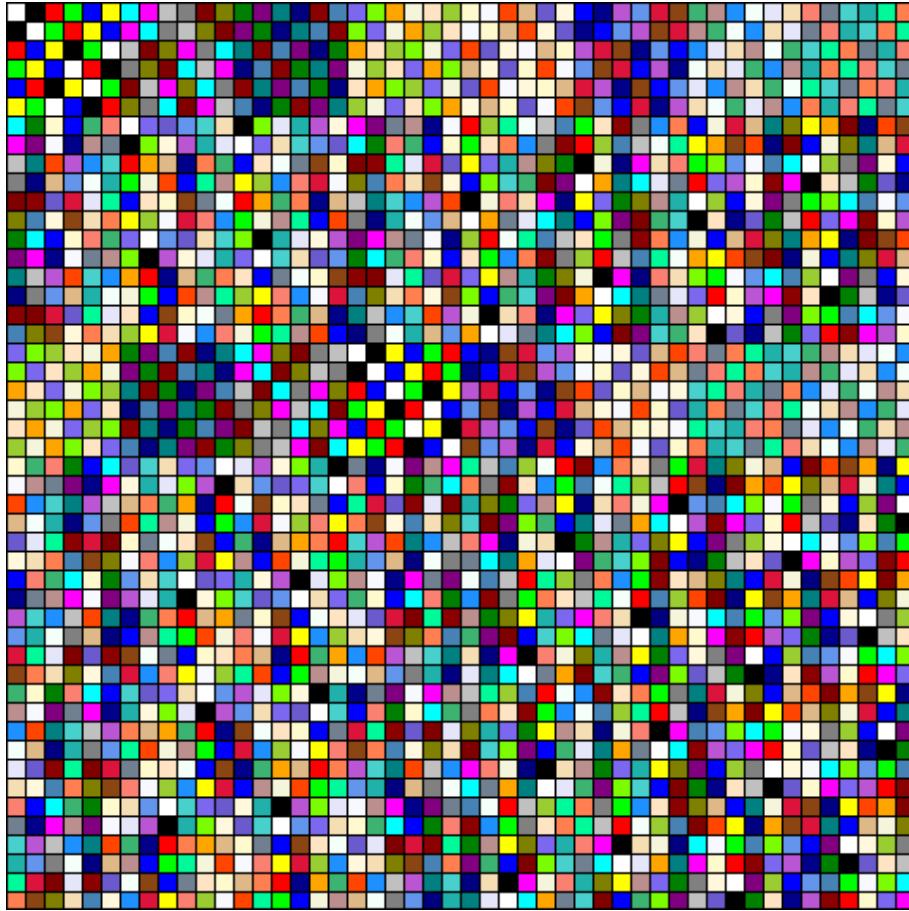
$$, \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$$

$$, \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix},$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \Bigg\}.$$

The group table is shown in Figure 1.2. The ordering of rows and columns is the same as the ordering of group elements in this example.

Figure 1.2: The group table of $GL(2, 3)$

Example

The group $GL(2, 3)$ has 8 conjugacy classes of elements:

0	1	1A	1	48	48	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
1	2	2A	1	48	48	$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$
2	2	2B	12	4	4	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
3	3	3A	8	6	12	$\begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}$
4	4	4A	6	8	16	$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$
5	6	6A	8	6	12	$\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$
6	8	8A	6	8	16	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$
7	8	8B	6	8	16	$\begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix}$

Example

The group $GL(2, 3)$ has 16 conjugacy classes of subgroups:

The group order is

$$48$$

There are 16 classes of subgroups

The following table shows the conjugacy classes of subgroups. The columns display: the number of the class, the order of the subgroups in the class, and the size of the class.

0	1	1	1
1	2	1	1, 2
2	2	12	1, 2
3	3	4	1, 3 ²
4	4	3	1, 2, 4 ²
5	4	6	1, 2 ³
6	6	4	1, 2 ³ , 3 ²
7	6	4	1, 2, 3 ² , 6 ²
8	6	4	1, 2 ³ , 3 ²
9	8	1	1, 2, 4 ⁶
10	8	3	1, 2, 4 ² , 8 ⁴
11	8	3	1, 2 ⁵ , 4 ²
12	12	4	1, 2 ⁷ , 3 ² , 6 ²
13	16	3	1, 2 ⁵ , 4 ⁶ , 8 ⁴
14	24	1	1, 2, 3 ⁸ , 4 ⁶ , 6 ⁸
15	48	1	1, 2 ¹³ , 3 ⁸ , 4 ⁶ , 6 ⁸ , 8 ¹²

The Subgroup Lattice

Let G be a group, and let $\mathcal{C}$ be the set of conjugacy classes of subgroups of G . Two classes C_1 and C_2 are related if there are two groups $G_1 \in C_1$ and $G_2 \in C_2$, such that $G_1 \leq G_2$. Two classes are immediately related if there is no third class C_3 in between the two. This means that there is no $G_1 \in C_1$, $G_2 \in C_2$, and $G_3 \in C_3$ such that $G_1 \leq G_2 \leq G_3$.

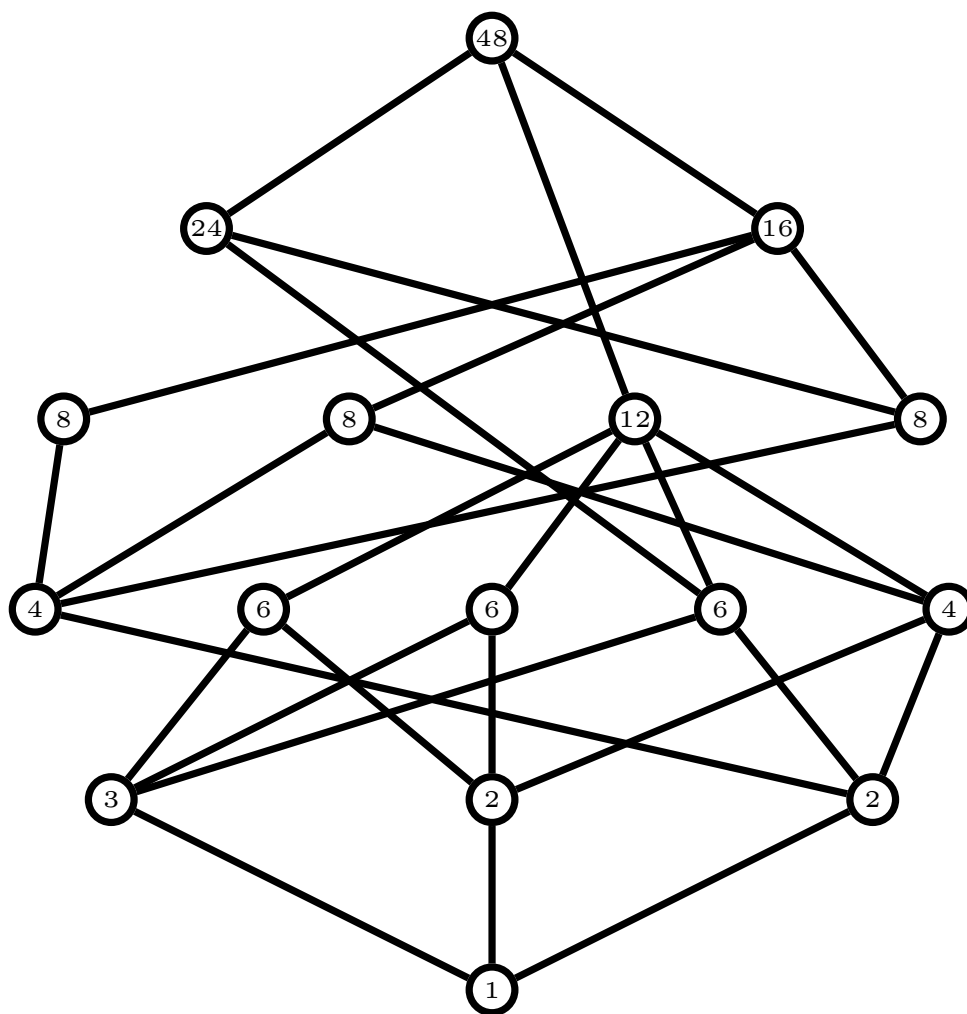
A drawing of the conjugacy classes is shown in Figure 1.3. The nodes represent conjugacy classes. The number in the node is the order of the groups in the class. The drawing in the figure shows the immediate relations between orbits of subgroups.

Comment

Let $|G| = n \cdot p^k$ where p is prime and k is maximal, i.e. $\gcd(n, p) = 1$. Any subgroup of order p^k of G is called p -Sylow subgroup of G (or Sylow p -subgroup). We see that $\text{GL}(2, 3)$ has one class of Sylow 3-subgroups of length 4 and one class of Sylow 2-subgroups of order 16 of length 3.

Comment

A group is simple if it does not have any normal subgroup apart from the trivial ones (the group itself and the identity subgroup). A subgroup of index 2 is always normal. As $\text{GL}(2, 3)$ has order 48 and contains a subgroup of order 24 (and index 2), the group is not simple.

Figure 1.3: The subgroup lattice of $GL(2, 3)$ by orbits

Assignment

Problem 1

Fill in the Theorem of Sylow:

Problem 2

Compute the order of $\text{GL}(n, q)$ for some small values of n and q . Make a table.

Problem 3

Let G be a group. Review the definition of the order of an element $g \in G$. Explain what the order

invariant of a subgroup is.

Problem 4

What is the permutation representation of $\mathrm{GL}(n, p)$?

Problem 5

Let G be a group. What is a generating set of G ? In the case of $\text{GL}(2, 3)$, what is a generating set?

The Projective General Linear Group – PGL

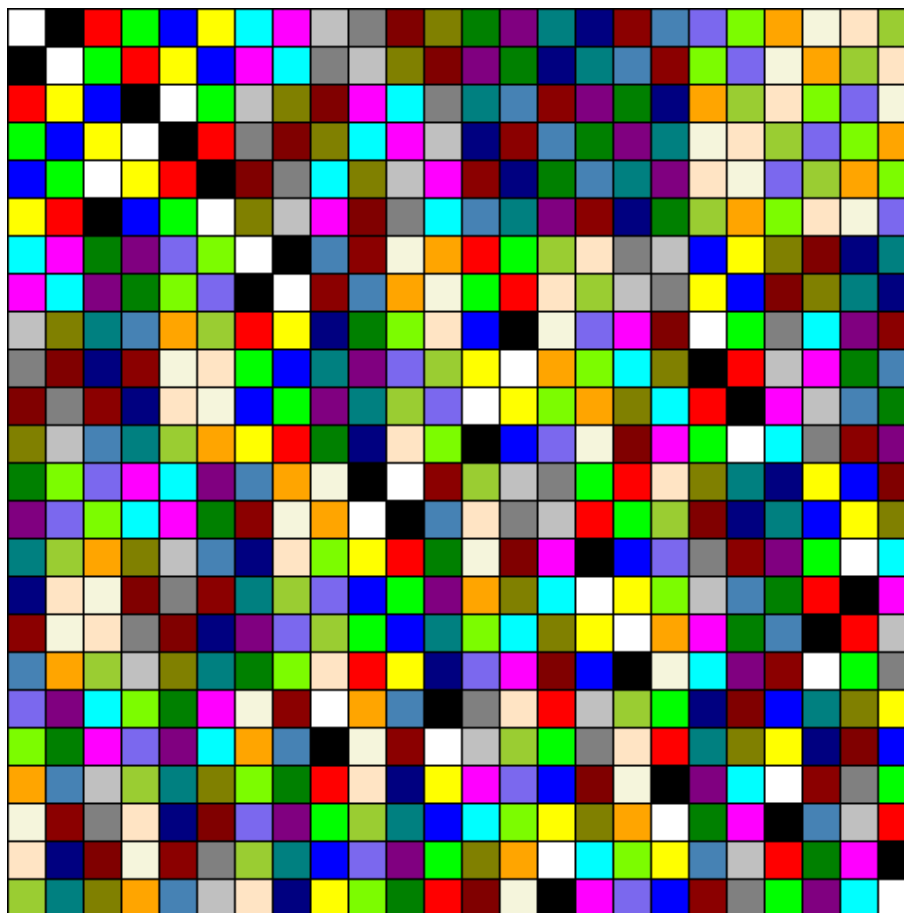
Let $G = \text{GL}(n, q)$ be the general linear group. Because $\Phi(c\mathbf{x}) = c\Phi(\mathbf{x})$, the group acts on one-dimensional subspaces. The kernel of the action is the group of diagonal matrices with one nonzero element on the diagonal. This subgroup is also the center of the group $\text{GL}(n, q)$. The factor group is called $\text{PGL}(n, q)$. The order of $\text{PGL}(n, q)$ is given by $|\text{GL}(n, q)|/(q - 1)$, as we have $q - 1$ elements in the center and each element of $\text{PGL}(n, q)$ is a coset of the center.

Example

The group $\text{PGL}(2, 3)$ has 24 ($= 48/2$) elements:

$$\begin{aligned} & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \\ & , \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \\ & \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

The group table is shown in Figure 1.4. The ordering of rows and columns is the same as the ordering of group elements in this example.

Figure 1.4: The group table of $\text{PGL}(2, 3)$